SrLab Title:- INVESTIGATING RHION POSSESSION EVIDENCE

Student Name:- BADAMASI MUNTAKA MUHAMMAD

Student ID:- 2025FWSD11563

Course Name:- COMPUTER FORENSICS CASE STUDIES

Instructor Name:- COMMANDANT AMINU IDRIS

Date Of Submission:- 26/10/2026

Case scenario

This is a fictional training investigation concerning imagery that may meet an illegal-possession threshold in the case scenario. The evidence package includes a seized USB disk image and network traces associated with FTP and HTTP activity. Investigators need to know whether deleted files, hidden data, reconstructed network objects, and the timing of recorded transfers form a defensible sequence. The task is an offline examination of supplied training artefacts; it does not authorise any contact with people, accounts, systems, hosts, or services named in the material.

Lab Objectives

The objectives of the lab is to understand how to collect digital evidences from different digital forensic aspect such as USB disk image and network traffic

---

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- wireshark:- this is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
- sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
- Md5sum:- Print/compute or check MD5 (128-bit) checksums message digest.
- xxd:- make a hexdump or do the reverse. creates a hex dump of a given file or standard input.
- bless:- this is graphical hexadecimal Gtk# editor
- unzip:- these tool list, test and extract compressed files in a ZIP archive
- fcrackzip:- this is a Free/Fast Zip Password Cracker. fcrackzip searches each zipfile given for encrypted files and tries to guess the password.

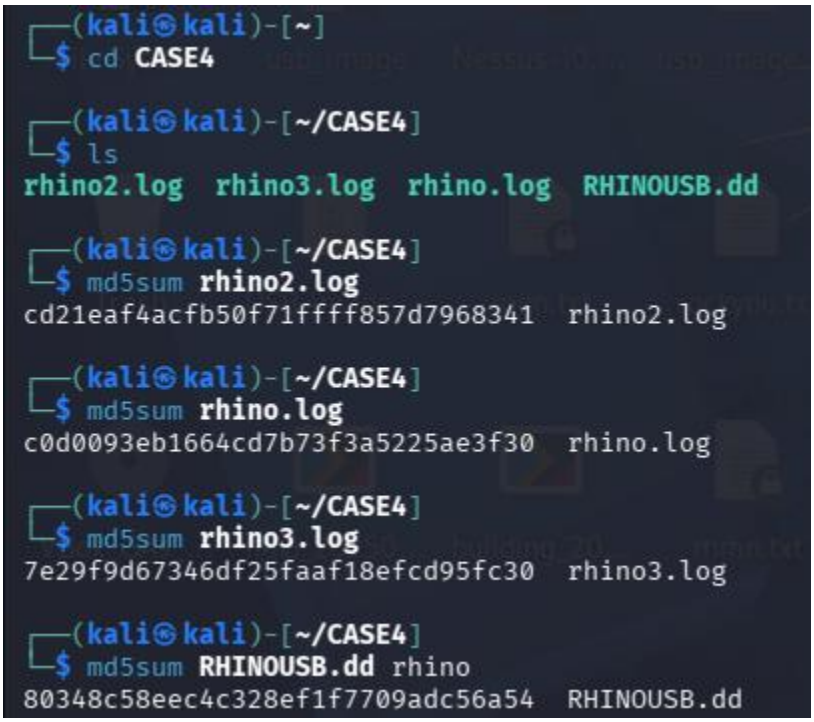
Methology

- I power on the kali linux machine
- I used the wget command to download the zip case study file(DFRWS2005-RODEO)
- I used the unzip command to unzip the case study file (DFRWS2005-RODEO)
- I obtain some basic information such as the MD5 & SHA256 hash values of the different file found inside the case study file (DFRWS2005-RODEO)
- I start analyzing the case study file (DFRWS2005-RODEO) using some digital forensic tools such as wireshark,exiftool,fcrackzip, etc

---

Screenshot and Evidence

SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE study file (DFRWS2005-RODEO)



```
┌──(root㉿kali)-[/home/kali/CASE4]
└─# mmls RHINOUSB.dd

┌──(root㉿kali)-[/home/kali/CASE4]
└─# fsstat RHINOUSB.dd
FILE SYSTEM INFORMATION
--------------------------------------------

File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
--------------------------------------------

Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)
--------------------------------------------
```

Use the command line

```
File Syste

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
--------------------------------------------

Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)
--------------------------------------------

529-536 (8) → EOF
537-544 (8) → EOF

┌──(root㉿kali)-[/home/kali/CASE4]
```

SCREENSHOT SHOWING SUCCESSFUL RECOVERY OF THE RECUP_DIR.1 FOLDER

```
┌──(kali㉿kali)-[~/CASE4]
└─$ cd recup_dir.1

┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$ ls
f0000545.txt                                  f0336969.txt  f0344681.txt  f0352393.txt  f0360105.txt  f0367817.txt  f0375529.txt  f0383241.txt  f0390945.txt
f0104057.jpg                                  f0337489.txt  f0345201.txt  f0352905.txt  f0360969.txt  f0368329.txt  f0376041.txt  f0383753.txt  f0391465.txt
f0104249.jpg                                  f0338001.txt  f0345713.txt  f0353425.txt  f0361137.txt  f0368841.txt  f0376553.txt  f0384265.txt  f0391977.txt
f0105065.jpg                                  f0338513.txt  f0346225.txt  f0353937.txt  f0361649.txt  f0369361.txt  f0377353.txt  f0384777.txt  f0392489.txt
f0105873.jpg                                  f0339025.txt  f0346737.txt  f0354449.txt  f0362161.txt  f0369873.txt  f0377585.txt  f0385297.txt  f0393001.txt
f0106393.jpg                                  f0339545.txt  f0347257.txt  f0354961.txt  f0362673.txt  f0370385.txt  f0378097.txt  f0385809.txt  f0393737.txt
f0106409.jpg                                  f0340057.txt  f0347769.txt  f0355481.txt  f0363193.txt  f0370897.txt  f0378609.txt  f0386321.txt  f0394033.txt
f0106865.gif                                  f0340569.txt  f0348281.txt  f0355993.txt  f0363705.txt  f0371417.txt  f0379129.txt  f0386833.txt  f0394545.txt
f0106889.gif                                  f0341081.txt  f0348793.txt  f0356505.txt  f0364217.txt  f0371929.txt  f0379641.txt  f0387353.txt  f0395065.txt
f0106905.txt                                  f0341601.txt  f0349313.txt  f0357017.txt  f0364729.txt  f0372441.txt  f0380153.txt  f0387865.txt  f0395577.txt
f0335017_She_died_in_February_at_the_age_of_74.doc  f0342113.txt  f0349825.txt  f0357537.txt  f0365249.txt  f0372953.txt  f0380665.txt  f0388377.txt  f0396089.txt
f0335081.jpg                                  f0342625.txt  f0350337.txt  f0358049.txt  f0365761.txt  f0373473.txt  f0381185.txt  f0388889.txt  f0396601.txt
f0335601.txt                                  f0343137.txt  f0350849.txt  f0358561.txt  f0366273.txt  f0373985.txt  f0381697.txt  f0389409.txt  report.xml
f0335945.txt                                  f0343657.txt  f0351369.txt  f0359073.txt  f0366785.txt  f0374497.txt  f0382209.txt  f0389921.txt
f0336457.txt                                  f0344585.txt  f0351881.txt  f0359593.txt  f0367305.txt  f0375009.txt  f0382721.txt  f0390433.txt

┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$ ls *.jpg
f0104057.jpg  f0104249.jpg  f0105065.jpg  f0105873.jpg  f0106393.jpg  f0106409.jpg  f0335081.jpg

┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$ ls -l *.jpg
-rw-r--r-- 1 root root  95814 Oct 21 10:15 f0104057.jpg
-rw-r--r-- 1 root root 415534 Oct 21 10:15 f0104249.jpg
-rw-r--r-- 1 root root 411361 Oct 21 10:15 f0105065.jpg
-rw-r--r-- 1 root root 264600 Oct 21 10:15 f0105873.jpg
-rw-r--r-- 1 root root   6809 Oct 21 10:15 f0106393.jpg
-rw-r--r-- 1 root root 230665 Oct 21 10:15 f0106409.jpg
-rw-r--r-- 1 root root 264600 Oct 21 10:15 f0335081.jpg

┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$
```

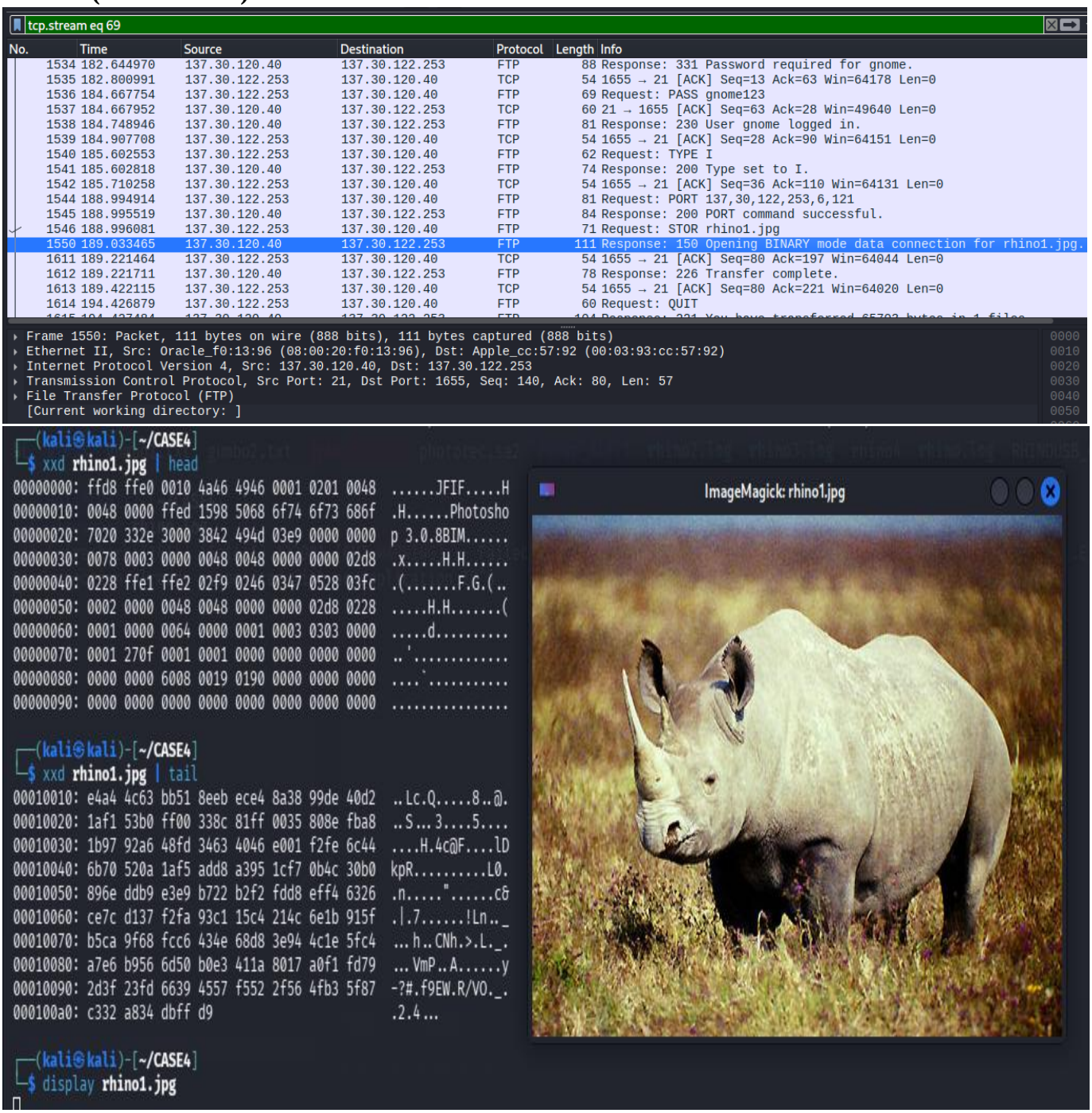
```
┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$ ls -l *.doc
-rw-r--r-- 1 root root 30720 Aug  8  2005 f0335017_She_died_in_February_at_the_age_of_74.doc

┌──(kali㉿kali)-[~/CASE4/recup_dir.1]
└─$
```

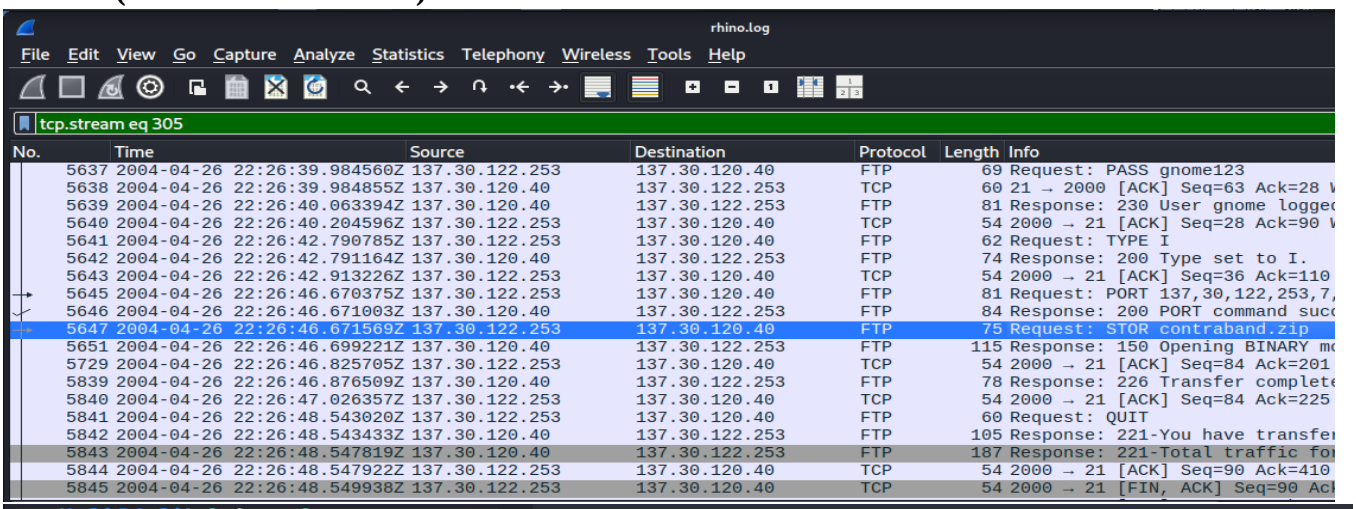
SCREENSHOT SHOWING ALL THE RECOVERED RHINO IMAGES FROM RECUP_DIR.1 FOLDER



SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(RHINO.JPG) FROM THE CAPTURED NETWORK TRAFFIC



SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(CONTRABAND.ZIP) FROM THE CAPTURED NETWORK TRAFFIC




```
┌──(kali㉿kali)-[~/CASE4]
└─$ ls
BL      contraband.zip  gumbo2.txt   photorec.se2  rhino1.jpg  rhino3.jpg  rhino4      RHINOUSB_CPY.dd  rockyou.txt
bless   gumbo1.txt      jphs_05.zip  recup_dir.1   rhino2.log  rhino3.log  rhino.log   RHINOUSB.dd

┌──(kali㉿kali)-[~/CASE4]
└─$ xxd contraband.zip | head
00000000: 504b 0304 1400 0100 0800 0688 9a30 65be  PK...........0e.
00000010: 6e93 3084 0300 0985 0300 0a00 0000 7268  n.0...........rh
00000020: 696e 6f32 2e6a 7067 c97e 43ad a105 2f94  ino2.jpg.~C.../.
00000030: cb18 8af6 1b1e f51e 89fb 6fcf 71c7 1e99  ..........o.q...
00000040: 69fa 9d19 7d17 2ebd 4b22 6f4c c482 2e2d  i...}...K"oL...-
00000050: d3b6 9559 fcaa b0a5 3315 bc7e cb63 510f  ...Y....3..~.cQ.
00000060: 6206 506d 1c0f e621 952c df42 b966 5918  b.Pm...!.,.B.fY.
00000070: 1b04 5956 bf7d e875 7d68 d9ac 84f8 bf1b  ..YV.}.u}h......
00000080: 257d 58e1 49fc a5b8 22eb 91d6 7da9 667f  %}X.I..."...}.f.
00000090: 330c ce4b 5ad8 3d8e c0ce 9002 cfe2 c954  3..KZ.=........T

┌──(kali㉿kali)-[~/CASE4]
└─$ xxd contraband.zip | tail
00038410: 9a1c f695 c479 a6f7 96f6 8c48 cecf 2469  .....y.....H..$i
00038420: f3fa 9467 1f96 ed27 8e90 d0e2 0f8e 11ea  ...g...'........
00038430: 9968 c966 5990 1c90 0e60 3391 de07 488d  .h.fY....`3...H.
00038440: 58cf 389d 9ca8 0d55 88ab 6660 3ea9 ce4e  X.8....U..f`>..N
00038450: 4df9 4279 9e23 d159 504b 0102 1400 1400  M.By.#.YPK......
00038460: 0100 0800 0688 9a30 65be 6e93 3084 0300  .......0e.n.0...
00038470: 0985 0300 0a00 0000 0000 0000 2000 0000  ............ ...
00038480: 0000 0000 0000 7268 696e 6f32 2e6a 7067  ......rhino2.jpg
00038490: 504b 0506 0000 0000 0100 0100 3800 0000  PK..........8...
000384a0: 5884 0300 0000                           X.....

┌──(kali㉿kali)-[~/CASE4]
└─$
```

```
┌──(kali㉿kali)-[~/CASE4]
└─$ unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:

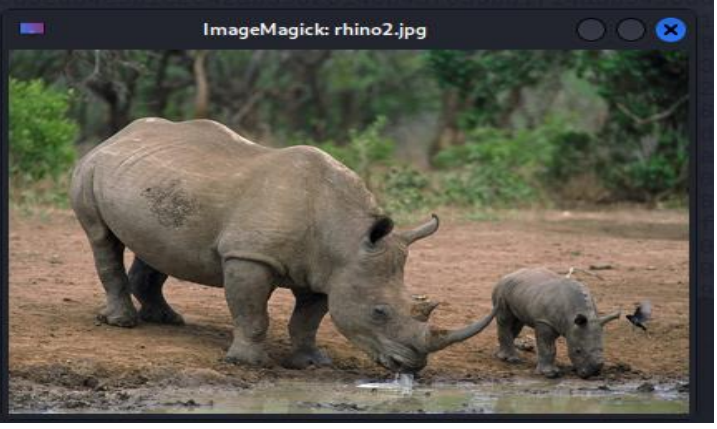
┌──(kali㉿kali)-[~/CASE4]
└─$ fcrackzip -u -D -p rockyou.txt contraband.zip


PASSWORD FOUND!!!!: pw == monkey

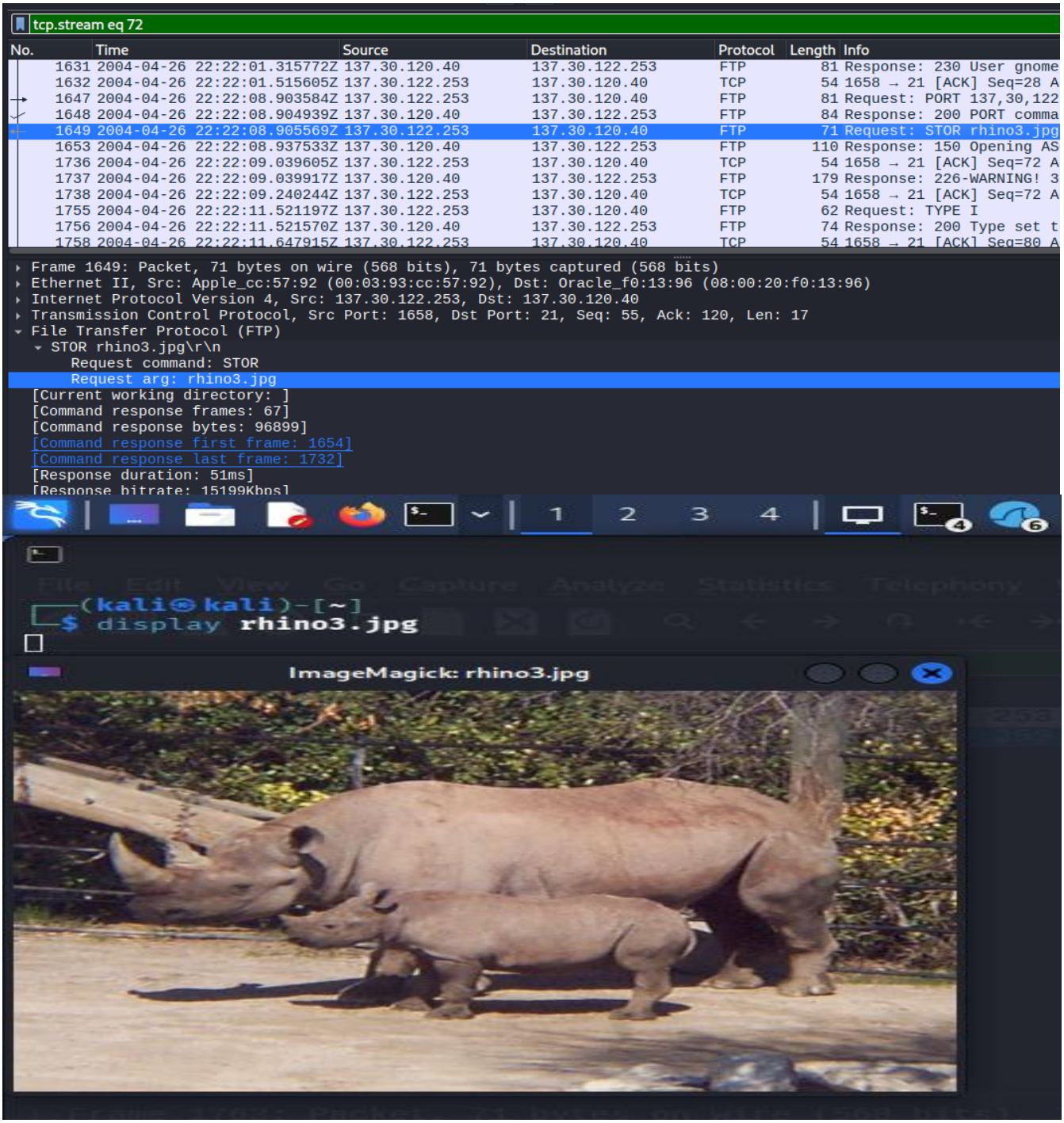
┌──(kali㉿kali)-[~/CASE4]
└─$ unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:
  inflating: rhino2.jpg

┌──(kali㉿kali)-[~/CASE4]
└─$ ls -l rhino2.jpg
-rw-rw-r-- 1 kali kali 230665 Apr 26  2004 rhino2.jpg

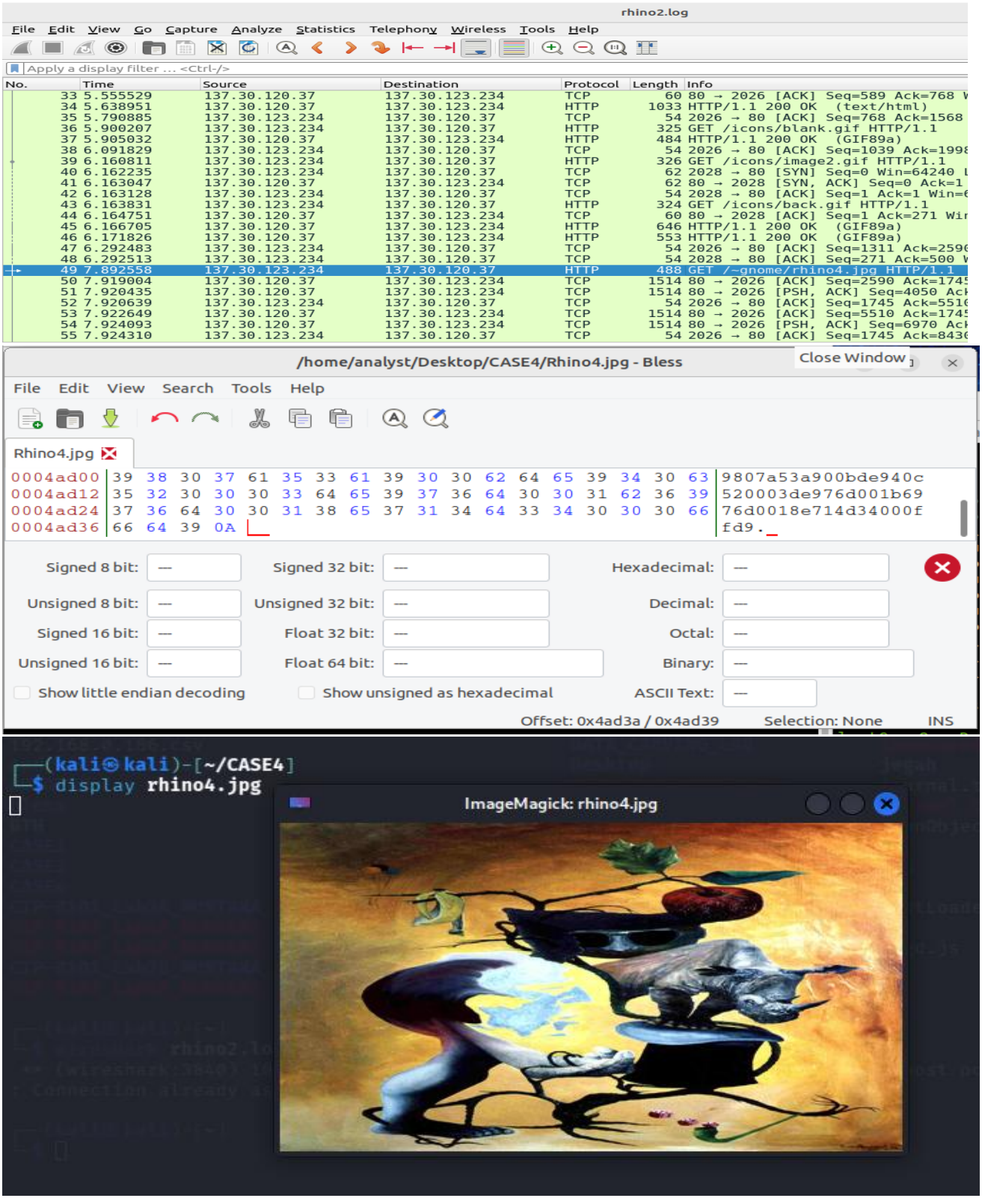
┌──(kali㉿kali)-[~/CASE4]
└─$ display rhino2.jpg
```



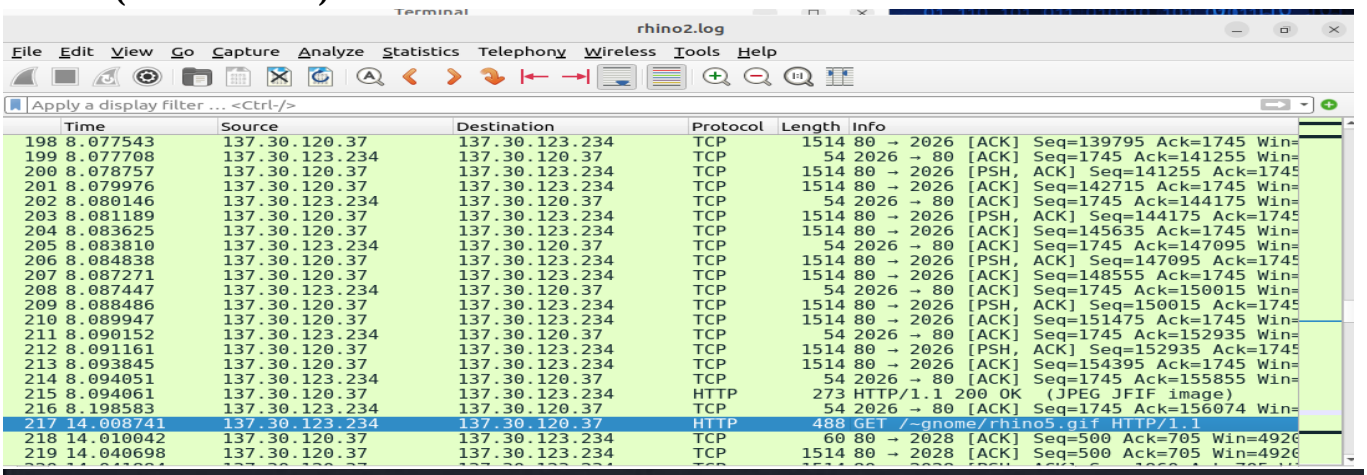
SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(RHINO3.JPG) FROM THE CAPTURED NETWORK TRAFFIC



SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(RHINO4.JPG) FROM THE CAPTURED NETWORK TRAFFIC

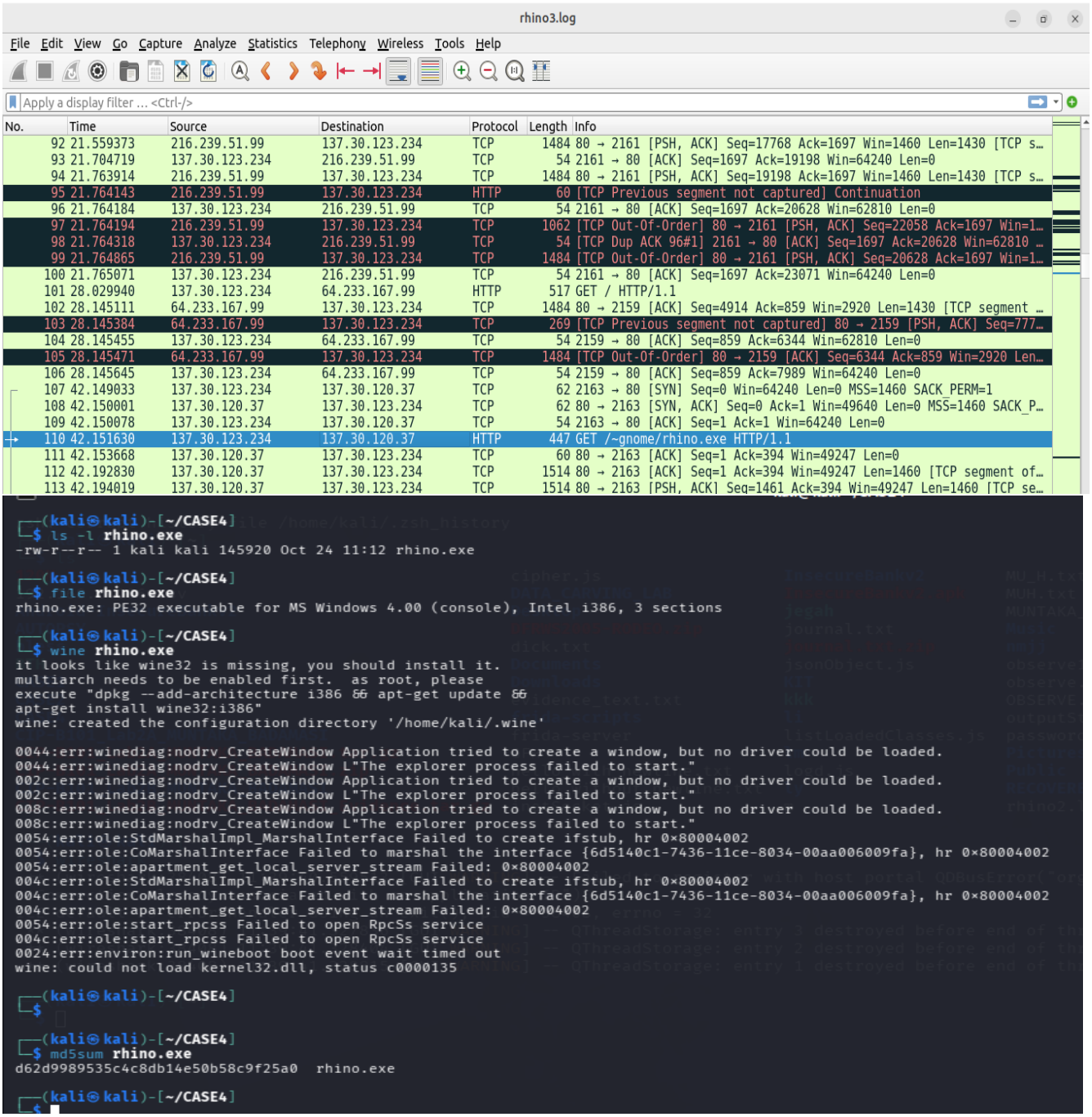


SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(RHINO5.JPG) FROM THE CAPTURED NETWORK TRAFFIC

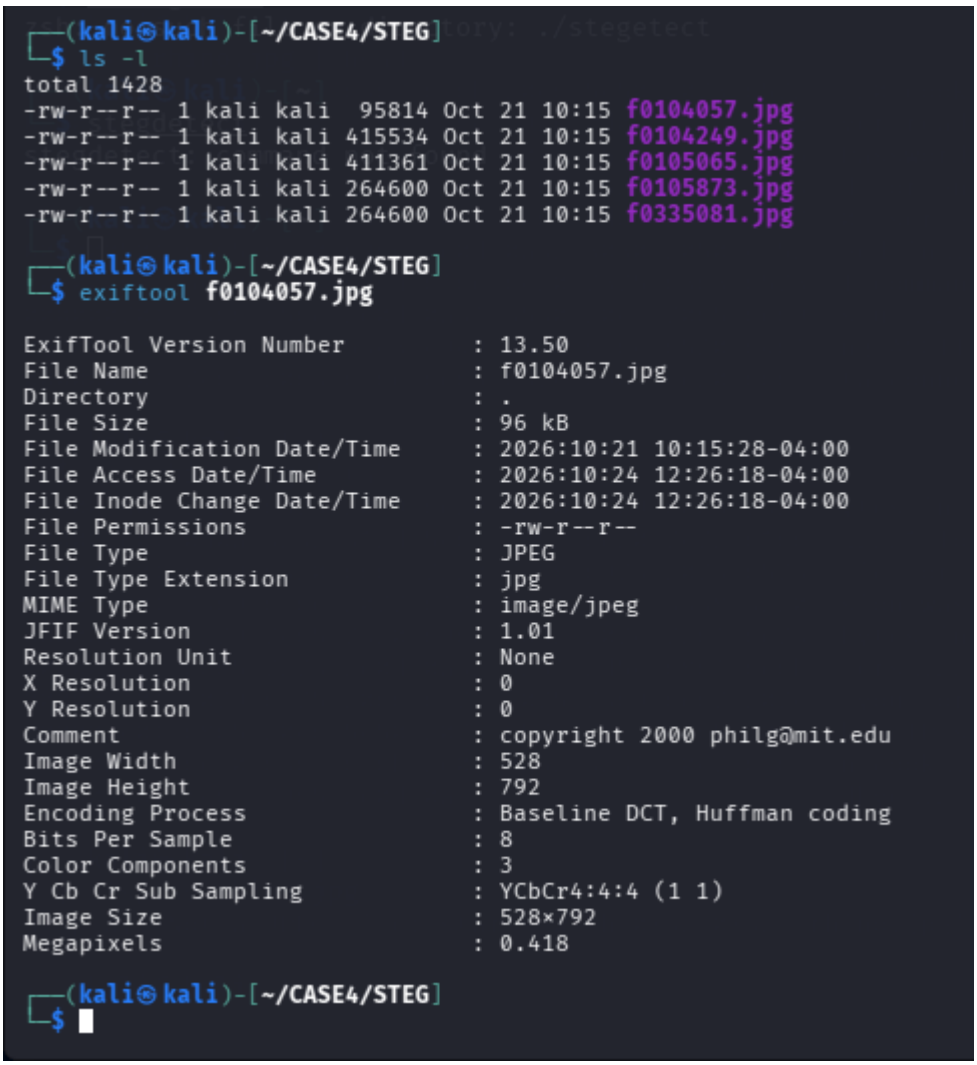


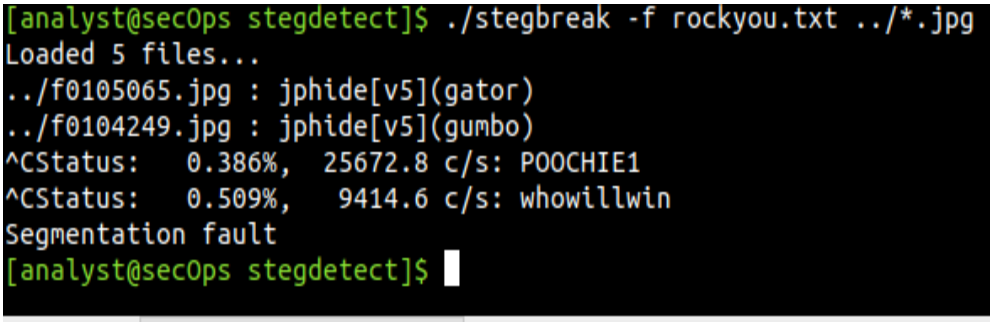


SCREENSHOT SHOWING THE SUCCESSFUL RECOVERY OF THE IMAGE(RHINO.EXE) FROM THE CAPTURED NETWORK TRAFFIC



SCREENSHOT SHOWING ANALYSING ALL THE NON-RHINO IMAGES





```
[analyst@secOps stegdetect]$ ./stegdetect
^C
[analyst@secOps stegdetect]$ ./stegdetect ~/home/analyst/STEG/* .jpg
/home/analyst/home/analyst/STEG/* : error: No such file or directory
.jpg : error: No such file or directory
[analyst@secOps stegdetect]$ ./stegdetect ~/STEG/* .jpg
/home/analyst/STEG/f0104057.jpg : negative
/home/analyst/STEG/f0104249.jpg : jphide(*)
/home/analyst/STEG/f0105065.jpg : skipped (false positive likely)
/home/analyst/STEG/f0105873.jpg : negative
/home/analyst/STEG/f0335081.jpg : negative
/home/analyst/STEG/stegdetect : error: Empty input file
.jpg : error: No such file or directory
[analyst@secOps stegdetect]$ ./stegdetect -s 3 ~/STEG/* .jpg
/home/analyst/STEG/f0104057.jpg : negative
/home/analyst/STEG/f0104249.jpg : jphide(***)
/home/analyst/STEG/f0105065.jpg : skipped (false positive likely)
/home/analyst/STEG/f0105873.jpg : jphide(*)
/home/analyst/STEG/f0335081.jpg : jphide(*)
/home/analyst/STEG/stegdetect : error: Empty input file
.jpg : error: No such file or directory
```

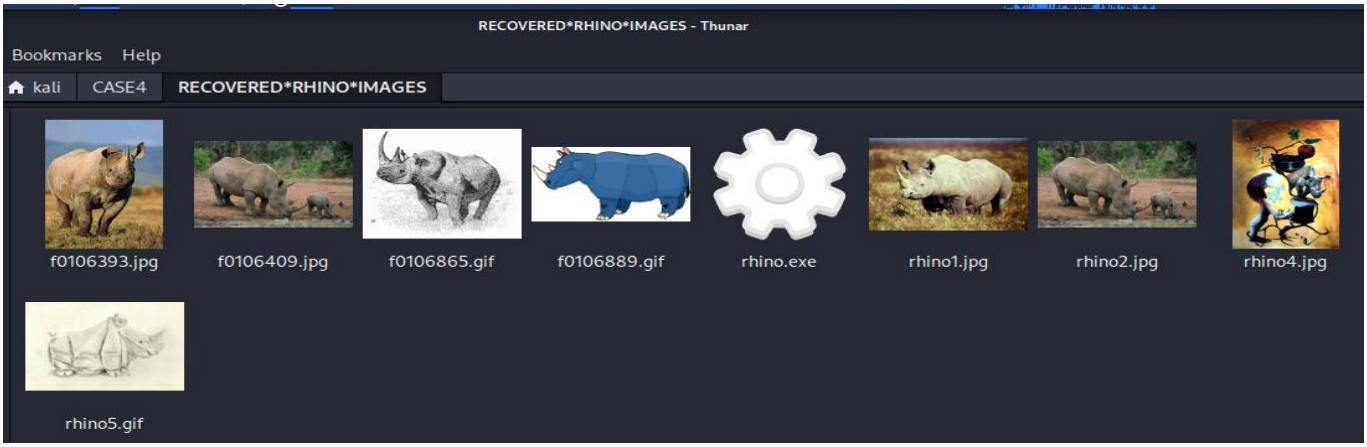
```
[analyst@secOps STEG]$ cd  stegseekexe/
[analyst@secOps stegseekexe]$ wget ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip
--2026-10-25 16:50:06--  ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip
           => ‘jphs_05.zip’
Resolving ftp.gwdg.de (ftp.gwdg.de)... 134.76.12.6, 2a06:93c7:f0::f79
Connecting to ftp.gwdg.de (ftp.gwdg.de)|134.76.12.6|:21... connected.
Logging in as anonymous ... Logged in!
==> SYST ... done.    ==> PWD ... done.
==> TYPE I ... done.  ==> CWD (1) /pub/linux/misc/ppdd ... done.
==> SIZE jphs_05.zip ... 184359
==> PASV ... done.    ==> RETR jphs_05.zip ... done.
Length: 184359 (180K) (unauthoritative)

jphs_05.zip                         100%[=================================================================>] 180.04K  37.0KB/s    in 24s

2026-10-25 16:50:42 (7.35 KB/s) - ‘jphs_05.zip’ saved [184359]

[analyst@secOps stegseekexe]$ ls
jphs_05.zip
[analyst@secOps stegseekexe]$ unzip jphs_05.zip
Archive:  jphs_05.zip
 extracting: jphs05.zip
 extracting: jphs05.zip.sig
[analyst@secOps stegseekexe]$ unzip jphs05.zip
Archive:  jphs05.zip
  inflating: jphide.exe
  inflating: jpseek.exe
  inflating: Jphswin.exe
  inflating: Readme.txt
```

SCREENSHOT SHOWING ALL THE RECOVERED RHINO IMAGES



Conclusion

In conclusion this lab has laid a very strong understanding on how to use recover evidences relating to images transferred over a network using tools like wireshark.

Reference

➢ICDFA course Material (SBT-DF204(computer forensic case studies) by COMMANDANT AMINU IDRIS
